

Gedächtnisprotokoll

Data Literacy

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Klausur:	Hauptklausur WS 21/22
Pruefer:	Prof. Hennig
Datum:	15.02.2017
Zeit:	90min
Punkte:	100
Hilfsmittel:	Taschenrechner
Sprache:	Englisch

1 Text comprehension and Bayes Theorem:

a)

Vaccine efficacy is defined as:

$$VE := \frac{p(h|\neg v) - p(h|v)}{p(h|\neg v)}$$

Given that 85% of people are vaccinated, find $p(v|h)$ including $p(v)$ and VE.

b)

Plug in values to get a numeric measure of the $p(v|h)$

c)

one value was adapted (booster leads to 90% efficacy). Calculate probability like in b) with adapted value.

2 Given iid samples $[x_1, x_2, \dots, x_n]$ from population(?)

a)

How are mean and variance defined?

b)

Give estimators for mean and variance.

c)

Show that estimators are unbiased.

3 Statistical Testing

How likely is that student cant participate in exam due to Covid? 7 day incidence for Tuebingen given. Binomial distribution given where f is probability to be infected. Plots of binomial distribution and survival function $s(c)$ were given. Black line in $s(c)$ showed a p value of 5%.

a)

How many students can't take the exam (most probable number)?

b)

Null hypothesis is rejected at $p < 5\%$. How many student got to be infected, so that the Null is rejected?

c)

Suppose 10 people cancel the exam due to a covid infection. Someone says: 'The Null is false. Some people are lying about their covid infection'. Name (at least) two thinks that are problematic in this assumption.

4 MLE estimation on Gamma distribution

a)

find the MLE estimator for θ .

b)

Define the Fisher information and using the result from above.

c)

Show the asymptotic normality of the MLE estimator and the variance of the Distribution. $\hat{\theta}$.

5 Regression: log likelihood of error function of ω (?) was given. Also data was visualized (looked like exponential function)

a)

Find the ω that minimizes the log likelihood

Why is a linear regression not appropriate to fit the data?

Give a function $z(x)$, that would be more suitable to fit the data.

b)

Give w_0 , w_1 and w_2 that would classify the distribution without mistake.

c)

Using linear combination of non-linear functions and Gaussian noise for the given data. Is this a good choice for a model?

6 Logistic regression: Visualized binary labeled dataset was given with dimension k_1 and k_2

$$s = \omega_0 + \omega_1 x_{k_1} + \omega_2 x_{k_2},$$
$$\text{and } \sigma(s) = \frac{1}{1 + \exp(-s)}$$

was given

a)

Give the negative Log likelihood and its derivative

b)

give a ω that has no classification error. Is this ω unique?

c)

?

7 Fairness: $P(R, A, Y)$ was given with concrete values for $P(R = 1, Y = y, A = a) = 0.xx$ etc.

a)

Give the marginal distribution $p(R)$

b)

Does this distribution fulfill the fairness criterion of separation? Why?

8 3 plots with different x, y scales were given. Define $f(x) = ?$ for each plot